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Advanced Geochemical Signatures for Monitoring CO2 Fate and Behavior

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Abstract

Carbon Capture and storage (CCS) technology represents a critical strategy for mitigating greenhouse gas emissions and achieving net-zero targets. Despite the promise of the various global technologies and tracer monitoring methods to evaluate the fate of injected CO_2 in subsurface environments, substantial obstacles remain, particularly in terms of cost, regulatory compliance, and the scalability of these solutions. This study presents an experimental geochemical investigation aimed at stimulating the injectivity and storage capabilities of CO_2. The approach was mixing two end-members at different rates; reservoir native CO_2 ($\delta^{13}C$ baseline measured average equal to −3.5‰) and injected CO_2 with $\delta^{13}C$ value equal to −10.1‰. The stable isotopic compositions of the gas samples were measured by GC-IRMS. International gas standards are injected prior to sample analysis to check accuracy, and the precision for daily runs must be ± 0.2‰ for carbon isotopes.

The results were obtained from 30 gas samples representing 3 well scenarios with variable CO_2 concentrations and compositions. The samples contain variable mixing rates of injected and native CO_2; therefore, they were divided into three groups as A, B, and C, based on their $\delta^{13}C$- CO_2 values. The average carbon isotopic compositions of CO_2 for group A, group B, and group C are −9.7‰, −4.5‰, and −2.8‰, respectively. Their associated average CO_2 concentration (mol%) for the three groups are 91%, 19%, and 46%, respectively. Mass-balance calculations were used to quantify the proportion of injected CO_2 contributed to the received samples, assuming no isotopic fractionation affected $\delta^{13}C$-CO_2. The produced gases from group A, group B, and group C contain 83.8%, 2.6%, and 7.95% of the injected CO_2, respectively. The findings of this study are pivotal for enhancing our understanding of optimal zones for CO_2 injectivity and storage. The findings indicate that carbon isotopic analysis is a robust method for monitoring the movement and fate of CO_2 in complex subsurface environments, providing valuable insights for CCS initiatives.

Introduction

Carbon capture and storage (CCS) is one of the most effective solutions for reducing greenhouse gas emissions into the atmosphere (IEA, 2009). These greenhouse gases play a major role in climate change and global warming. To mitigate the effect of greenhouse gases and to meet global energy demand that is predicted to increase by 47% due to the growth of economy and population, various measures are considered to reduce carbon emissions (Davoodi et al, 2024). For instance, increase energy efficiency, rely more on renewable energy sources, and increase usage of electric cars (Blesl et al, 2007). In addition, CO_2 sequestration and CO_2-enhanced oil recovery (CO_2-EOR) are successful means to achieve these goals. After capturing CO_2 from the atmosphere, it is either transported away or stored back in a selected depleted gas and oil reservoirs or saline aquifers using the solubility trapping mechanism (Rezk et al, 2019). The stored CO_2 requires monitoring and tracing, which can be done by multiple geophysical and geochemical approaches. For more accurate geochemical analysis, it is required to regularly collect fluid samples from the observation wells (Mayer et al, 2013). The injected CO_2 will react with reservoir rocks, resulting in petrophysical, geochemical and geo-mechanical alterations that affect the macroscopic CO_2 dynamics in the reservoir (Kharaka and Cole, 2011; Myrttinen et al., 2012; Mayer et al., 2013; Cui et al., 2021; Famoori et al., 2021; Ozotta et al., 2021). In addition, injected CO_2 interacts with resident formation water (brine) and dissolves in the water. This will result in the formation of carbonic acid then dissociates into bicarbonate (HCO_3^-) and protons (H^+), consequently lowering the pH values. This process is known as solubility trapping of injected CO_2 (Mayer et al, 2013).

$$CO_2 + H_2O \leftrightarrow \quad H_2CO_3 \leftrightarrow \quad H^+ + HCO_3^- \tag{1}$$

Another process called ionic trapping of the injected CO_2 as HCO_3^-, shown in equation 2. This occurs in calcite or dolomite reservoirs,

$$H_2O + CO_2 + CaCO^3 \leftrightarrow Ca^{2+} + 2HCO_3^- \tag{2}$$

Both reactions reduce the volume of buoyant CO_2, therefore increase storage capabilities (Mayer et al, 2013). The degree of CO_2 solubility depends on multiple geochemical parameters including pH, concentration of dissolved inorganic carbon (DIC), alkalinity, Ca, Mg, Fe, Cl, and Na (Mayer et al, 2013).

The interaction with formation water disturbs the acid-base equilibrium and triggers the dissolution of carbonate rocks and possibly the precipitation of minerals (Gharbi et al., 2013; Grigg and Svec, 2006; Gunter et al., 1997; Zhang et al., 2017). Examples of minerals that can be dissolved during CO_2 flooding are carbonate minerals (dominantly calcite) and aluminium silicate (mainly feldspar) (Song et al, 2019). pH values could be lowered to 3 and growth of potassium crystals when CO_2 injected into sandstone as the case in Illionois Basin in the United States (Song et al, 2019). Permeability increases with mineral dissolution but decreases with precipitation of new minerals, such as clay minerals that cause clogging migration pathways (Song et al, 2019). Several monitoring techniques were tested worldwide at various scales to assess the behaviour and fate of injected CO_2 in the subsurface, including chemical and gas tracers (e.g., Myers et al., 2013; Roberts et al., 2017), gas composition (e.g., Boreham et al., 2011), and isotopes (e.g., Johnson et al., 2011; Karolytė et al., 2019). Examples of these traces are Kr and SF6, which were added to the CO_2 injected stream. The downside of most geochemical traces is their huge cost in large quantities, as millions of tons of CO_2 are injected over many years or decades (Mayer et al, 2013).

Stable isotopes geochemical analysis is more powerful tool when the objective is to monitor the reactions of CO_2 in the subsurface and/or to differentiate injected CO_2 from the formation native CO_2 derived from host rocks and hydrocarbon gases. They are cost-effective with no additional cost needed. This method is mostly effective when there is a large difference between the indigenous and injected CO_2 (Mayer et al, 2013).

Geological storage of CO_2 is a key component in the global effort on Carbon Capture & Sequestration (CCS) to reduce greenhouse gas emission rates. One of the most volumetrically significant underground storage options is to inject CO_2 into a mature hydrocarbon field gas or oil, to enhance oil recovery. The experiment was designed to understand the behavior of the injected CO_2 using both compositional and isotopic fingerprinting techniques for selected gas samples.

Methods

30 gas samples were collected in pressurized 500 cc gas cylinders and sent to GU for geochemical analysis. Gas composition analysis was conducted using gas chromatography by the Hydrocarbon Phase and Behaviour Unit (HPBU). The stable isotopic composition of the gas sample components is measured by GC-IRMS (gas chromatography–isotope ratio mass spectrometry). The gas samples are injected into a gas chromatograph (GC IsoLink, Thermo) where hydrocarbon compounds are separated according to their size and polarity. For carbon isotope analysis, each hydrocarbon chromatographic peak is converted into CO_2 and H2O by a combustion reactor at 1020°C. H_2O is trapped and CO_2 is carried by helium to the source of the mass spectrometer where it is ionized.

Ions formed in the IRMS (Delta V Plus, Thermo) source (CO_2^+, H_2^+) are deflected by a magnet according to their mass and collected by Faraday cups. The software calculates the ratio of the heavy to the light isotope and provides the isotopic composition as a delta notation in per mil vs reference standard.

Instrument calibration is performed using international standards from IAEA and internal standards after any maintenance event or major change in the set-up. A daily calibration check is performed using commercial and in-house isotope standards at the beginning, during, and at the end of each batch of samples.

Standards are injected prior to sample analysis to check accuracy and repeatability. The precision for daily runs must be ± 0.2‰ for carbon isotopes to validate the batch. Within one batch, at least one sample is run in duplicate.

Measurements are reported in delta notation as part-per-thousand (‰) variations in atomic $^{13}C/^{12}C$ relative to the VPDB international reporting standard:

$$\delta^{13}C(‰) = 1000*(R_{SAMPLE}/R_{VPDB} - 1), \text{ where } R = {}^{13}C/{}^{12}C \tag{3}$$

Negative $\delta^{13}C$ values indicate enrichment of ^{12}C relative to the VPDB Standard, whereas positive values indicate enrichment of ^{13}C.

Results and discussion

Carbon isotopes of CO_2 from 30 gas samples range from −10.52 to −2.3‰. The lowest carbon isotope value is equal to −10.52‰, and the highest is equal to −2.3‰. To assess the origin of the natural gases associated with the CO_2, a plot of methane $\delta^{13}C$ versus gas wetness (C^{2+}hydrocarbons / total hydrocarbon gases) was used Figure 1. It is modified from Bernard et al. (1976), Schoell (1983), and Clayton (1991). All the studied gases are plotted in the region of oil-associated thermogenic origin (Figure 1), but group C gases are drier compared to group A and B gases. The carbon isotopic profiles from C1-C3 and CO_2 of the studied gases were plotted in Figure 2. From the plot, the studied gases have almost identical C1-C3 isotopic values, suggesting they originated from the same source rock. However, there is a clear difference in CO_2 values that divides the data into three distinct groups. The variations in CO_2 values indicate an ununited source for this non-hydrocarbon gas.

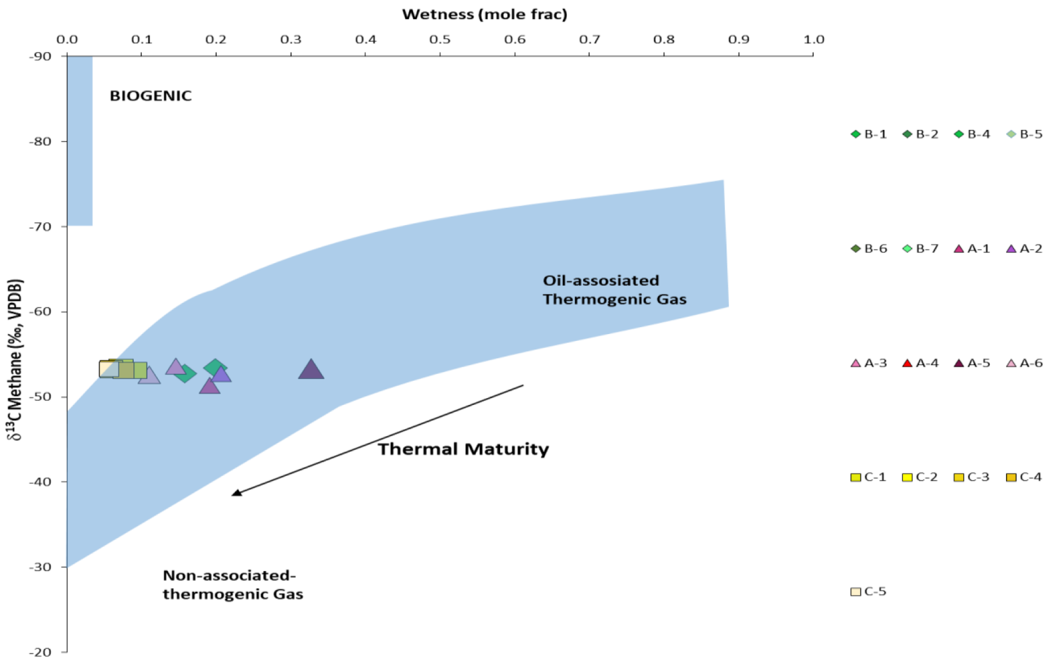

Figure 1—Plot of Carbon isotopic composition of methane ($^{13}\delta CCH4$) versus wetness (C2+/Total HC Gas) of the studied gases. Shows all the selected gas samples are oil-associated thermogenic origin.

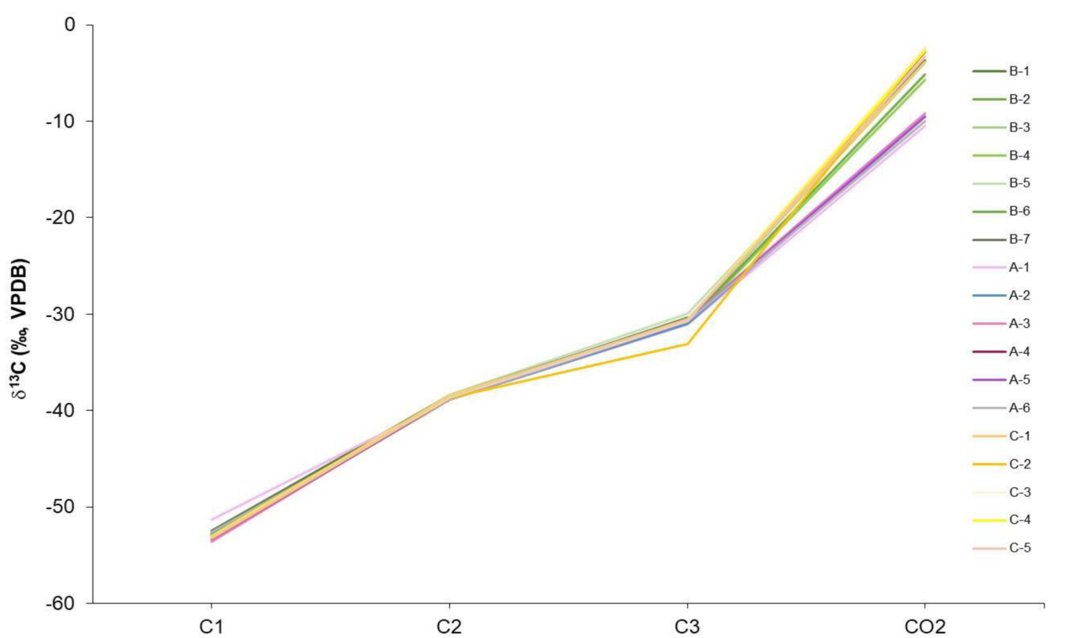

Figure 2—Carbon isotopic composition profiles of C1-C3 and CO_2.

Reference native CO_2 (baseline) has isotopic compositions ranging from −6.0 to 1.6‰, with a measured average equal to −3.5‰, on the other hand, carbon isotopes of the injection CO_2 are around −10.1‰ and assumed concentration 100%, which are shown in Figure 3. Isotopic gas compositions from groups B and C fall within the range observed for the indigenous CO_2 (Jenden and Al-Harthi, 2013; Lu et al., 2021) in Figure 3. However, group A isotopic signatures fall outside the range of the indigenous CO_2 values, indicating a different source of CO_2. Figure 4 displays $\delta^{13}C$-CO_2 versus the concentration of CO_2 (mol%) of the studied gas samples. This plot shows a similar pattern to Figure 3, where groups B and C gases are clustered close to native CO_2. Meanwhile, group A gas samples have $\delta^{13}C$ CO_2 values close to −10‰ and concentrations close to 100 mol%. An exception of group A gases is sample A3 which is plotted near groups B and C gas samples. Figure 5 displays $\delta^{13}C$-CO_2 ‰ versus depth (ft). The red vertical lines represent the $\delta^{13}C$ CO_2 values of injected CO_2 and the measured average indigenous CO_2 values. The isotopic values of group A gases increase as the depth increases, suggesting more indigenous (reservoir CO_2) signatures with increasing depth.

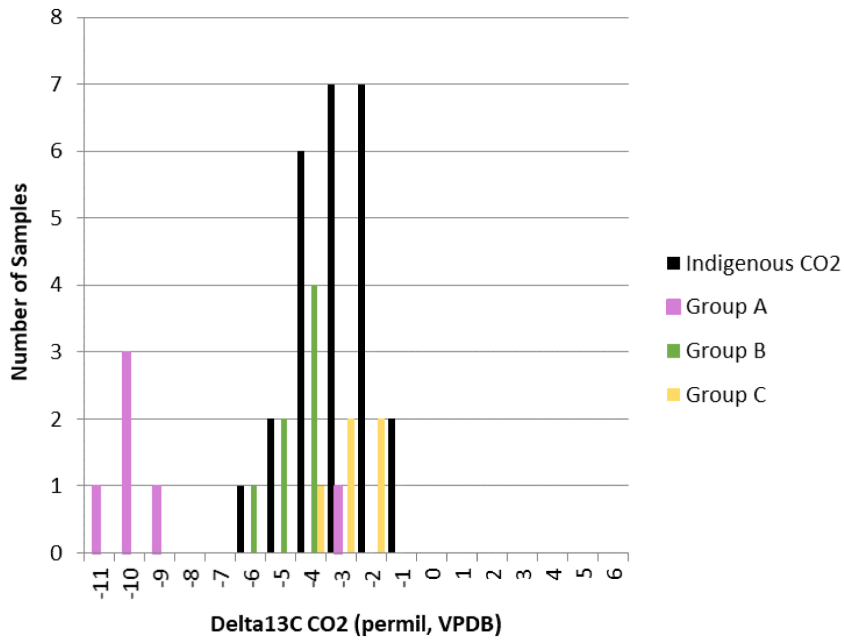

Figure 3—Range of $\delta^{13}C$-CO_2 values for Indigenous CO_2 and gas samples.

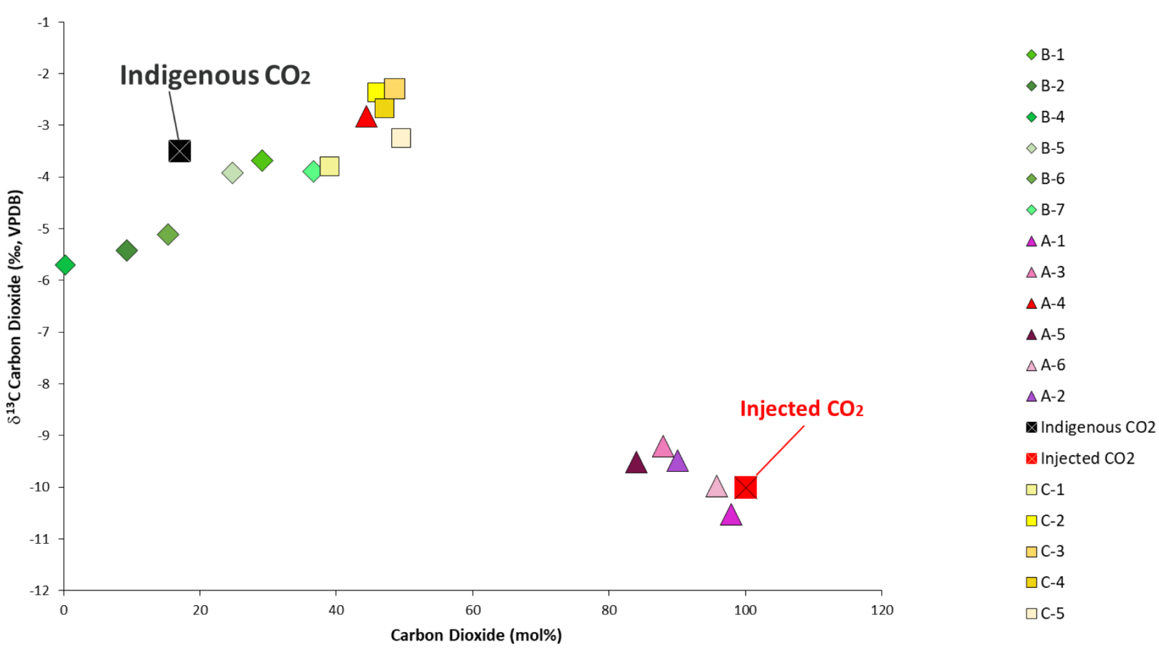

Figure 4—Carbon isotopic composition of CO_2 (‰, VPDB) versus the concentration of CO_2 (mol%).

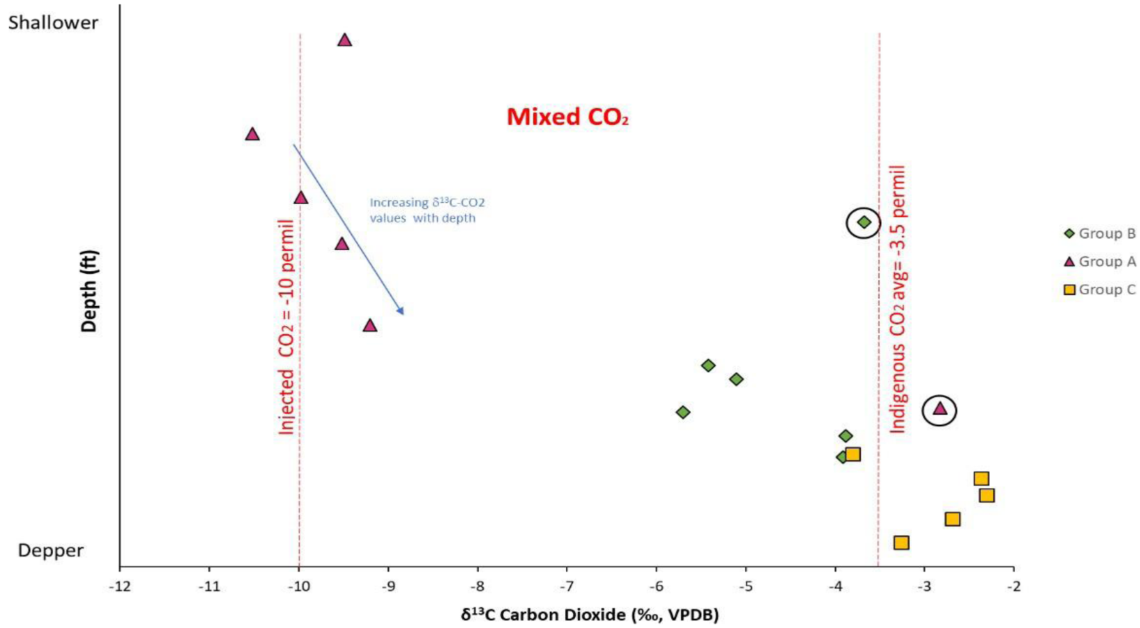

Figure 5—Carbon δ^{13}C CO_2 (V-PDB ‰) VS depth (ft). the red vertical lines represent the δ^{13}C CO_2 values of injected CO_2 and the measured average indigenous CO_2 values

Both the chemical composition and carbon stable isotopes of all end members was used to quantify/calculate how much of the injected CO_2 and original indigenous CO_2 using the below equation (Mayer et al, 2015)

$$\delta^{13}C_{mix} \times m_{mix} = \delta^{13}C_{bl} \times m_{bl} \times (1 - X) + \delta^{13}C_{inj} \times m_{inj} \times X \tag{4}$$

The $\delta^{13}C$ and m indicate C isotope ratios and concentrations of injected (inj) and baseline (bl) inorganic carbon and (mix) is the resulting mixtures. The fraction of injected volume in the final mixture is X. As the difference between the carbon isotopic compositions of indigenous native CO_2 and injected CO_2 is large, carbon isotope measurements of CO_2 from producing wells may be useful for monitoring the progress of the CO_2 flood front.

Isotopic mass-balance calculations using the equation (4) indicate that the produced CO_2 in groups A, B, and C contain 88%, 2.9%, and 7.8% of the injected CO_2, respectively. As illustrated in Figure 6, the bubble size represents the average percentage of injected CO_2 in each sample. In Figure 6, the percentage of injected CO_2 was calculated for each individual sample in the three groups and was plotted against sample depth. Group A gas samples contain the highest percentages of injected CO_2 and were collected from the shallowest intervals. On the other end, groups B and C gas samples have relatively close percentages of injected CO_2 values despite the clear difference in their isotopic compositions. This is due to the higher concentration of CO_2 in group C gas samples compared to group B gases as shown in Figure 4.

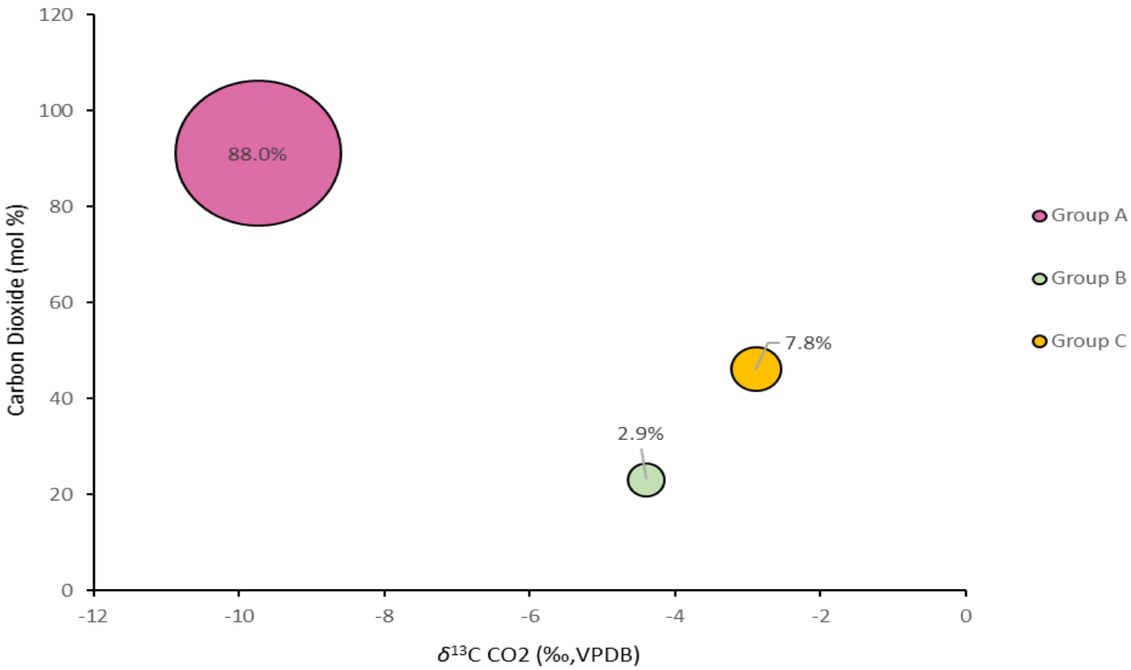

Figure 6—Plot of Carbon isotopic composition of CO_2 (‰, VPDB) versus concentration of CO_2 (mol%) and the bubble size represent the average percentage of injected CO_2 in each group.

The mass-balance results were computed on the assumption that the isotopic value obtained from the indigenous native CO_2 is −3.5‰ and the mol concentration is equal to 17%. More accurate results of the percentage of injected CO_2 versus formation CO_2 can be obtained from measuring the isotopic value of the formation using an offset well from the CO_2injection area to minimize the variation in the carbon isotopic signatures resulted from formation rock heterogeneity.

Conclusions and recommendations

Based on isotopic mass-balance, the results suggest different CO_2 emplacements in the reservoir. In addition, the results from this study illustrate that carbon isotopic composition can be used to trace the injected CO_2 in depleted reservoirs and determine the origin. We conclude that monitoring the movement of CO_2 in the injection reservoir with geochemical and isotopic techniques is an effective approach to determine plume expansion and to identify potential preferential flow paths, provided that the isotopic composition of injected CO_2 is constant and distinct from that of baseline CO_2. Given the breadth of the applications of carbon isotopes for natural gas. It is highly recommended for future applications that the CO_2 isotope assessment is periodically deployed (e.g., quarterly or semi-annually) on multiple samples collected from the different wells in the pilot area and offset wells to effectively monitor CO_2 emplacement in any selected fields.

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